ASSIGNMENT 1 – CO1 | TASK 3

Data Types and Type Compatibility

Scenario: Namma Metro Smart-Card Fare System

Objective / explanation

Money is stored in integer paise using long long. This avoids decimal rounding and provides enough range for large network totals.

C++ source code

```
#include &lt;climits&gt;
#include &lt;iostream&gt;
using namespace std;

int main() {
    long long journeys = 3000000LL;
    long long totalPaise = journeys * 3000LL;
    cout &lt;&lt; "sizeof(int): " &lt;&lt; sizeof(int) &lt;&lt; '\n';
    cout &lt;&lt; "INT_MAX: " &lt;&lt; INT_MAX &lt;&lt; '\n';
    cout &lt;&lt; "Total paise: " &lt;&lt; totalPaise &lt;&lt; '\n';
    cout &lt;&lt; "Total rupees: " &lt;&lt; totalPaise / 100 &lt;&lt; '\n';
}
```

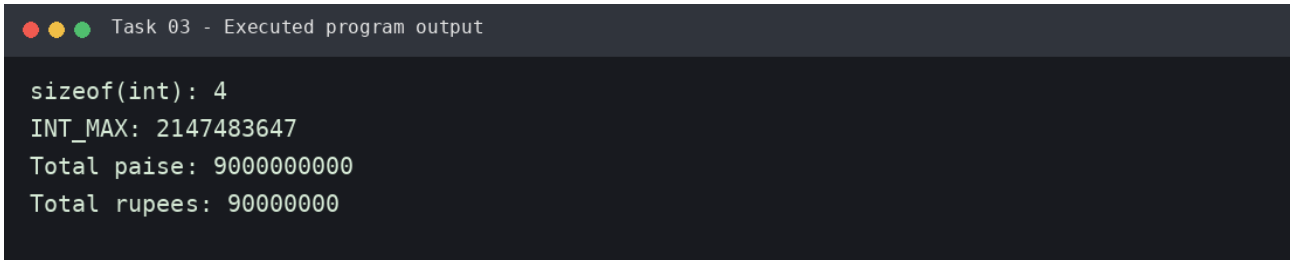
Compilation

```
g++ -std=c++17 -Wall -Wextra -pedantic main.cpp -o program
```

Actual captured program output

```
sizeof(int): 4
INT_MAX: 2147483647
Total paise: 9000000000
Total rupees: 90000000
```

Output screenshot



Analysis / complexity

3,000,000 × 3,000 paise = 9,000,000,000 paise, exceeding the usual 32-bit signed int maximum. Use long long for accumulated totals. sizeof output depends on the compiler/platform.

Evidence note: Output shown is captured by executing this source in the generation environment. Recompile and run on your own system, and use your own screenshots if your instructor requires evidence from your computer.